

命题演算形式系统

2201101

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一、公理

公理

公理集合：

(1) A_1: $A \to (B \to A)$

(2) A_2: $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$

(3) A_3: $(\neg A \to \neg B) \to (B \to A)$

推理规则或分离规则（Modus Ponens）：

若有A和$A \to B$成立，则必有结论B成立，可形式化表示为:

$$r_{mp}: \frac{A,\ A \to B}{B}$$



二、基本定理

1 基本定理

定理1：$\vdash_{PC} A \to A$ （$A \to A$是PC中的一个定理）√

定理2：如果$\vdash_{PC} A \to (B \to C)$，那么$\vdash_{PC} B \to (A \to C)$ （前件互换定理）√

定理3：$\vdash (A \to (B \to C)) \to (B \to (A \to C))$ 定理（2）的另一种形式 √

定理4：$\vdash (B \to C) \to ((A \to B) \to (A \to C))$（加前件定理）√

定理5：$\vdash (A \to B) \to ((B \to C) \to (A \to C))$（加后件定理）√

定理6：$\vdash \neg A \to (A \to B)$√

定理7：$\vdash A \to (\neg A \to B)$√

定理8：如果$\vdash (A \to B)$，$\vdash (B \to C)$，那么$\vdash (A \to C)$（三段论定理）√

定理9. $\vdash (\neg A \to A) \to A$ （反证法）√

定理10. $\vdash \neg\neg A \to A$√

定理11. $\vdash (A \to \neg A) \to \neg A$ （反证法）√

定理12. $\vdash A \to \neg\neg A$√





定理13：$\vdash (A \to B) \to (\neg B \to \neg A)$ **（公理A_3的逆命题）** √

定理14：$\vdash (\neg A \to B) \to (\neg B \to A)$ √

定理15：$\vdash (A \to \neg B) \to (B \to \neg A)$ √

定理16：$\vdash (\neg A \to B) \to ((\neg A \to \neg B) \to A)$ **（反证法）** √

定理17：$\vdash (A \to B) \to ((A \to \neg B) \to \neg A)$√

定理18：$\vdash \neg A \to C$ ，$\vdash B \to C$当且仅当 $\vdash (A \to B) \to C$√

定理19：$\vdash A \to A \vee B$， 其中，$A \vee B$定义为$\neg A \to B$，也即√

$$A \to A \vee B \Leftrightarrow A \to (\neg A \to B)$$ **（等价于定理7）**

定理20：$\vdash A \to B \vee A$， 其中，$A \vee B$定义为$\neg A \to B$，也即√

$$A \to B \vee A \Leftrightarrow A \to (\neg B \to A)$$ **（等价于公理1）**

定理21：如果$\vdash P \to Q$, 且$\vdash R \to S$, 则$\vdash (Q \to R) \to (P \to S)$√

定理22：$\vdash (A \to C) \to ((B \to C) \to ((A \vee B) \to C))$ 也即√

$$(A \to C) \to ((B \to C) \to ((\neg A \to B) \to C))$$ **（二难推理）**



三、主要解题手段

逆否公式群

公理3: $(\neg A \to \neg B) \to (B \to A)$

定理13: $\vdash (A \to B) \to (\neg B \to \neg A)$ **(公理A_3的逆命题)**

定理14: $\vdash (\neg A \to B) \to (\neg B \to A)$

定理15: $\vdash (A \to \neg B) \to (B \to \neg A)$

逆否公式群是十分常用的手段，在推理中运用可以得到相同的前件或后件。

2 定理18

定理18：$\vdash \neg A \to C$ ，　$\vdash B \to C$当且仅当 $\vdash (A \to B) \to C$√

定理18是一个相当强大的解题工具，如果看到前面的式子是可以拆分的形式，可以考虑使用定理18

3 前件互换定理

定理2：如果 $\vdash_{PC} A \to (B \to C)$，那么 $\vdash_{PC} B \to (A \to C)$ **（前件互换定理）** √

定理3：$\vdash (A \to (B \to C)) \to (B \to (A \to C))$ **定理（2）的另一种形式** √

公式的推理证明的重中之重在于找到相同的前件或者后件，提取出来从而化简公式，因此，定理2，定理3的重要性不言而喻。



四、例题讲解

(1) $(A \to (B \to C)) \to ((C \to D) \to (A \to (B \to D)))$

⇑ Thm 2

$(C \to D) \to ((A \to (B \to C)) \to (A \to (B \to D)))$

⇑ Thm 4

$(C \to D) \to ((B \to C) \to (B \to D))$

⇑ Thm 4

$(C \to D) \to [C \to D]$ Thm 1

(2) $((A \to B) \to (A \to C)) \to (A \to (B \to C))$

Thm18

$(A \to C) \to (A \to (B \to C))$

Thm4

$C \to (B \to C)$ A_1

$\neg(A \to B) \to (A \to (B \to C))$

?

$A \to (A \to (B \to C))$

$\neg B \to (A \to (B \to C))$

Thm2

$A \to (\neg B \to (B \to C))$ Thm6

$(\neg B \to (B \to C)) \to (A \to (\neg B \to (B \to C)))$ A_1

A_1, MP

$\neg(A \to B) \to ?$

$A \to ?$

$\neg B \to ?$

$(\neg(A \to B) \to A) \to (\neg A \to (A \to B))$ Thm 13 $\Leftarrow$ Thm 6

$(\neg(A \to B) \to \neg B) \to (B \to (A \to B))$ A3 $\Leftarrow$ A1

(3) $(A \to C) \to ((B \to C) \to ((\neg A \to B) \to C))$

Thm 18

$\neg A \to (\neg B \to C) \to ((\neg A \to B) \to C)$

↑ Thm 5

$\neg A \to ((\neg A \to B) \to B)$

↑ Thm 2

$(\neg A \to B) \to (\neg A \to B)$ Thm 1

$C \to (\neg B \to C) \to ((\neg A \to B) \to C)$

A_1

↑ Thm 2

$\neg B \to C \to (C \to (\neg A \to B) \to C)$ A_1

A_1, MP

(1) $(A \to (A \to B)) \to (A \to B)$

由 A_2

$A \to (A \to B) \to B)$

由 Thm 2

$(A \to B) \to (A \to B)$ Thm 1

注2: ① $\to A \to (A \to B)$ Thm 6　　② $(A \to B) \to (A \to B)$ Thm 1

(4) $(A \to (B \to C)) \to ((A \to (D \to B)) \to (A \to (D \to C)))$

Thm 18

A_2、Thm 4

① $A \to (A \to (D \to B)) \to (A \to (D \to C))$

A

$\uparrow A_2$

$\to A \to (A \to (D \to B)) \to (D \to C)$

Thm 6

② $(B \to C) \to ((A \to (D \to B)) \to (A \to (D \to C)))$

$\uparrow$ Thm 4

$(B \to C) \to ((D \to B) \to (D \to C))$

$\uparrow$ Thm 4

$(B \to C) \to (B \to C)$

Thm 1

(5) $((A \to (B \to C)) \to ((C \to D) \to (A \to (B \to D))))$

↑ Thm 2

$(C \to D) \to ((A \to (B \to C)) \to (A \to (B \to D)))$

↑ Thm 4

$(C \to D) \to ((B \to C) \to (B \to D))$

↑ Thm 4

$(C \to D) \to (C \to D)$ Thm 1

法2：$(A\to(B\to C))\to((C\to D)\to(A\to(B\to D)))$

Thm18

① $\to A\to((C\to D)\to(A\to(B\to D)))$

Thm2

pmp

Thm6

$(C\to D)\to(\to A\to(A\to(B\to D)))$

A1 , pmp

② $(B\to C)\to((C\to D)\to(A\to(B\to D)))$

Thm2

$(B\to C)\to(A\to((C\to D)\to(B\to D)))$

Thm5

$(B\to C)\to(A\to(B\to C))$

A1

(6) $((A \to B) \to C) \to (B \to C)$

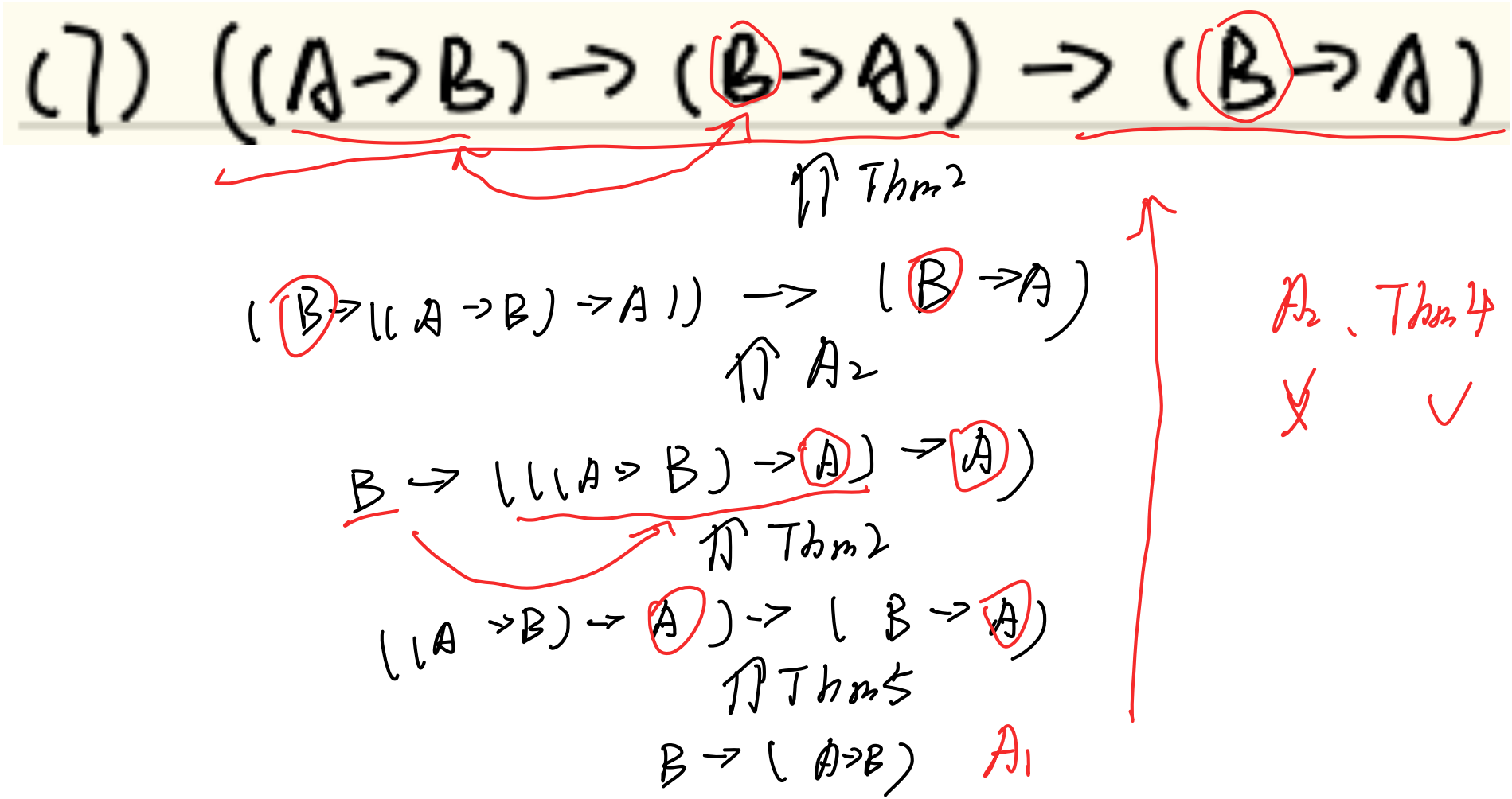
用 Thm 5

$B \to (A \to B)$ A1

① $\neg(A \to B) \to (B \to C)$ ✓

$\begin{cases} A \to (B \to C) \\ \neg B \to (B \to C) \end{cases}$ ✓ Thm 6

② $C \to (B \to C)$ A1


(7) ((A→B)→(B→A))→(B→A)
由 Thm 2
(B→(¬A→B)→A))→(B→A)
由 A2
B→(¬(¬A→B)→A)→A)
由 Thm 2
(¬A→B)→A)→(B→A)
由 Thm 3
B→(A→B) A1
A2、Thm 4
✗ ✓

法2： $(\neg A \to B) \to (B \to A) \to (B \to A)$

Thm 18 ?

① $\neg(A \to B) \to (B \to A)$

$$\begin{cases} A \to (B \to A) & A_1 \\ \neg B \to (B \to A) & \text{Thm 6} \end{cases}$$

② $(B \to A) \to (B \to A)$ Thm 1

(8) $A \to ((A \to B) \to (C \to B))$

介Thm5

$A \to (C \to A)$ A1

(9) $((A \to B) \Rightarrow A) \to A$

Thm 18

① $A \to A$ Thm 1

① $\to (A \to B) \to A$

↑ Thm 14

$\neg A \to (A \to B)$ Thm 6

(10) $((A \to B) \to C) \to ((C \to A) \to A)$

Thm 18

① $\neg(A \to B) \to ((C \to A) \to A)$

$A \to ((C \to A) \to A)$ ✓

$\neg B \to ((C \to A) \to B)$

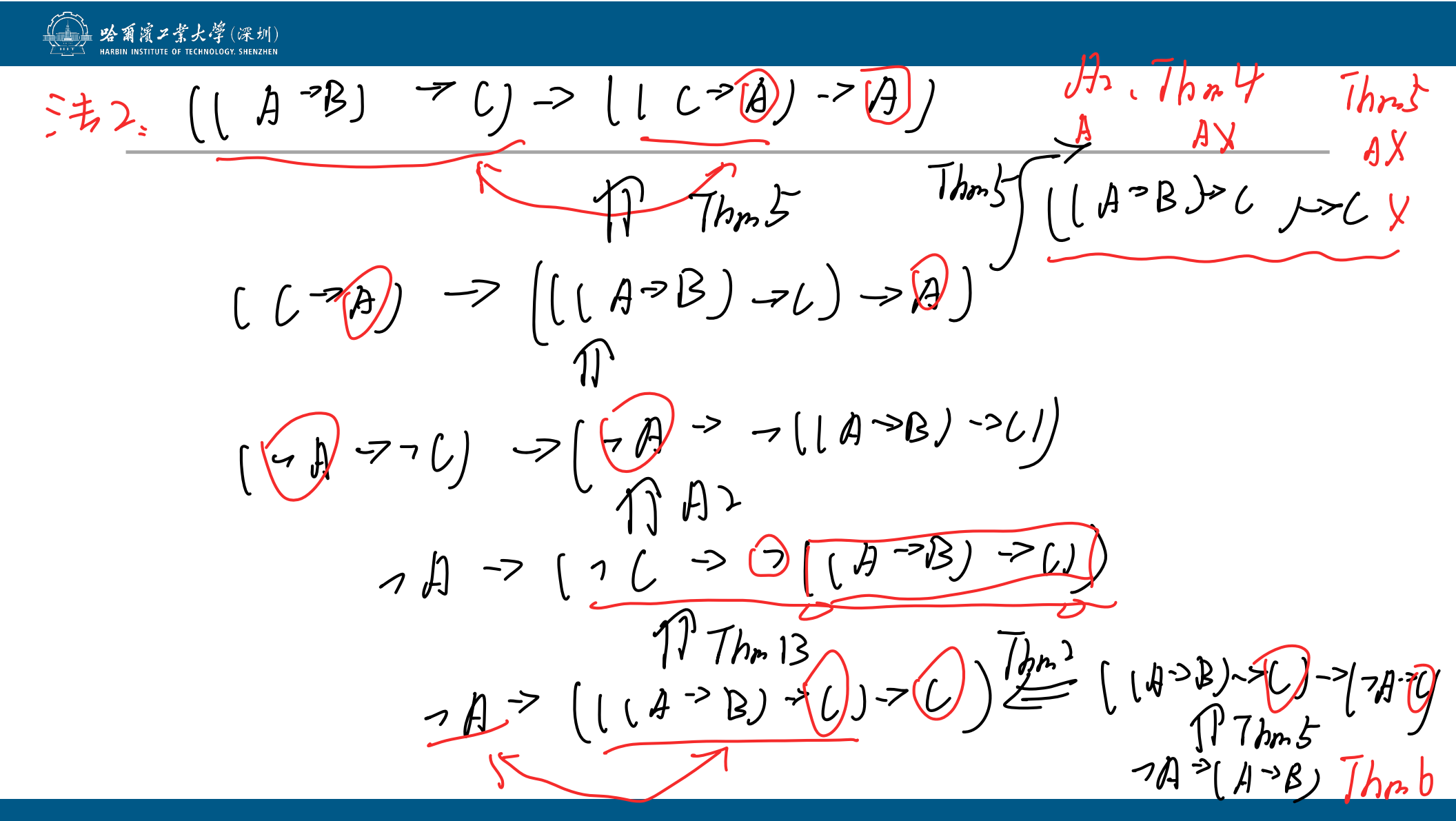
$(C \to A) \to (C \to A)$ Thm 1

Thm 2

② $C \to ((C \to A) \to A)$

由 Thm 2

$(C \to A) \to (C \to A)$ Thm 1


法2:
A2, Thm4
Thm5
Thm5
Thm5
Thm13
Thm2
Thm5
Thm6

$(A\to C)\to(A\to B)$ ✗

(11) $((A\to B)\to C)\to((A\to C)\to C)$

⇑ A3, Thm13

$(\neg C\to\neg(A\to B))\to(\neg C\to\neg(A\to C))$

⇑ A2

$\neg C\to(\neg(A\to B)\to\neg(A\to C))$

⇑ Thm13

$\neg C\to((A\to C)\to(A\to B))$

⇑ Thm 4

$\neg C\to(C\to B)$ Thm 6

法2: $((A\to B)\to C)\to((A\to C)\to C)$

Thm18

① $\neg(A\to B)\to(\neg A\to C)\to C)$ ✓

$\begin{cases} A\to(\neg A\to C)\to C) \\ \neg B\to(\neg A\to C)\to C) \end{cases}$ Thm2

$(A\to C)\to(A\to C)$ Thm1

② $C\to(\neg A\to C)\to C)$ A1

(12) $((((A\to B)\to C)\to D)\to((B\to D)\to(A\to D))$

Thm 18

① $\neg((A\to B)\to C)\to((B\to D)\to(A\to D))$

$(A\to B)\to((B\to D)\to(A\to D))$ ✓

$\neg C\to((B\to D)\to(A\to D))$

$(A\to B)\to(A\to B)$ Thm 1

Thm 5

② $D\to((B\to D)\to(A\to D))$

Thm 2

$(B\to D)\to(D\to(A\to D))$ A_1

A_1, MP

法2：

$$((A\to B)\to\bot)\to D)\to(B\to D))\to(A\to D))$$

$\Uparrow A_3$

$$(\neg D\to\neg((A\to B)\to\bot))\to(\neg D\to\neg B)\to(\neg D\to\neg A))$$

$\Uparrow A_2$

$$(\neg D\to\neg((A\to B)\to\bot))\to(\neg D\to(\neg B\to\neg A))$$

$\Uparrow$ Thm 4

$$\neg((A\to B)\to\bot)\to(\neg B\to\neg A)$$

$$\begin{cases}(A\to B)\to(\neg B\to\neg A) & \checkmark\ \text{Thm 13}\\ \neg\bot\to(\neg B\to\neg A)\end{cases}$$

(13) $(A \to C) \to ((B \to C) \to ((A \to B) \to B) \to C))$

Thm 18

① $\to A \to (\to B \to C) \to (\to(\to A \to B) \to B) \to C)$

↑ Thm 5

$\to A \to (\to(\to A \to B) \to B) \to B)$

↑ Thm 2

$(\to A \to B) \to B) \to (\to A \to B)$

↑ Thm 5

$\to A \to (A \to B)$ Thm 6

② $C \to (\to B \to C) \to (\to(\to A \to B) \to B) \to C)$

↑ Thm 2

$(B \to C) \to (C \to (\to(A \to B) \to B) \to C)$

A_1

A_1, R mp

法2：$(A \to C) \to (B \to C) \to (((A \to B) \to B) \to C)$

由 Thm 13, A_2, A_3

$(\neg C \to \neg A) \to (\neg C \to (((A \to B) \to B) \to B))$

由 Thm 4

$\neg A \to (((A \to B) \to B) \to B)$

由 Thm 2

$((A \to B) \to B) \to (\neg A \to B)$

由 Thm 5

$\neg A \to (A \to B)$ Thm 6

(14) $(A \to C) \to ((B \to C) \to (((B \to A) \to A) \to C))$

Thm 18

① $\to A \to (\neg B \to \bot) \to (((\neg B \to A) \to A) \to C)$

↑ Thm 5

$\to A \to (\neg\neg B \to A) \to A) \to B)$

↑ Thm 2

$(\neg B \to A) \to A) \to (\neg A \to B)$

↑ Thm 15

$(\neg B \to A) \to A) \to (\neg B \to A)$

↑ Thm 5 $\to B \to (\neg B \to A)$ Thm 6

② $C \to ((\neg B \to \bot) \to (((\neg B \to A) \to A) \to C))$

↑ Thm 2

$[\neg B \to \bot] \to (C \to (((\neg B \to A) \to A) \to C))$ A_1

A_1, RMP

法2: $(A\to C)\to((B\to C)\to((B\to A)\to A)\to C))$

↑ Thm 13 A_2 A_3

$((\neg C\to\neg A)\to(\neg C\to((B\to A)\to A)\to B))$

↑ Thm 4

$\neg A\to((((B\to A)\to A)\to B)$

↑ Thm 2

$((B\to A)\to A)\to(\neg A\to B)$

↑ Thm 14

$((B\to A)\to A)\to(\neg B\to A)$

↑ Thm 5 $\neg B\to(B\to A)$ Thm 6

(1) $(B \to A) \to (\neg A \to \neg B)$

Thm 18

① $\neg B \to (\neg A \to \neg B)$ A1

② $A \to (\neg A \to \neg B)$ Thm 7

(2) $(A \to B) \to ((B \to C) \to (A \to C))$

↑ Thm 2

$(B \to C) \to ((A \to B) \to (A \to C))$

↑ Thm 4

$(B \to C) \to (B \to C)$ Thm 1

(4) $\neg(A \to B) \to (B \to A)$

$\vdash \begin{cases} A \to (B \to A) & A_1 \\ \neg B \to (B \to A) & \text{Thm } b \end{cases}$

证明：① $A \to (B \to A)$ 公理1

② $\neg A \to (A \to B)$ 定理b

③ $(\neg A \to (A \to B)) \to (\neg(A \to B) \to A)$ 定理4

④ $\neg(A \to B) \to A$ ②③使用MP分离规则

⑤ $\neg(A \to B) \to (B \to A)$ ④①使用三段论

(1) $(\neg A \to \neg B) \to ((\neg A \to B) \to A)$

Thm 18

① $\neg A \to ((\neg A \to B) \to A)$

Thm 10

$A \to ((\neg A \to B) \to A)$

A1

② $\neg B \to ((\neg A \to B) \to A)$

Thm 2

$(\neg A \to B) \to (\neg B \to A)$

Thm 14





谢谢，敬请指正！